

Secondary School E-Learning Platform

Discussion Document - For Review Before Development Begins

This document is meant to be read and discussed together before any development starts. It lays out the full shape of the project as currently understood, including open questions that still need to be talked through.

Background

This project began as a university-focused E-Learning Management System ("Redesign and Implementation of E-Learning Management System"), covering course registration, attendance tracking, automated emails, a quiz module, digital library, past questions repository, assignment module, virtual classes, and institutional email auto-generation. That system was hosted on Vercel, Render, and MongoDB Atlas.

We are now extending the concept to serve secondary schools as well. This is a deliberate shift in direction: rather than building for one specific school, the secondary school version is being designed as a multi-tenant platform that any secondary school can sign up for and use independently, on shared infrastructure.

What Carries Over vs. What's New

This is one of the first things worth discussing together: how much of the university system gets reused as-is, adapted, or rebuilt from scratch for the secondary school version.

From University System	Likely Status for Secondary School Version
Course registration	Replaced - secondary schools use class/arm placement, not course registration.
Attendance tracking	Reusable concept, adapted to class-based attendance.
Automated emails	Reusable - same underlying mechanism, new triggers/templates.
Quiz module	Possibly reusable with adaptation - to be discussed.
Digital library	Possibly reusable with adaptation - to be discussed.
Past questions repository	Very relevant for secondary schools (WAEC/NECO/exam prep) - worth prioritizing.
Assignment module	Reusable concept, adapted to class/subject structure.
Virtual classes	To be discussed - is this a priority for secondary schools at launch?
Institutional email auto-generation	To be discussed - needed per school, or optional?

Platform Model: Multi-Tenant, Self-Service

Any secondary school will be able to register on the platform and begin using it immediately, without needing manual approval. Each school's data (classes, subjects, teachers, students, parents, results) is kept fully isolated from every other school on the platform.

- Self-service signup - a school registers itself and is instantly active.
- Each school is assigned a unique school identifier used to scope all of its data.
- Email verification confirms the registering contact before first login.
- The first person to register a school automatically becomes that school's Admin.
- No cross-school visibility - one school can never see another school's data.

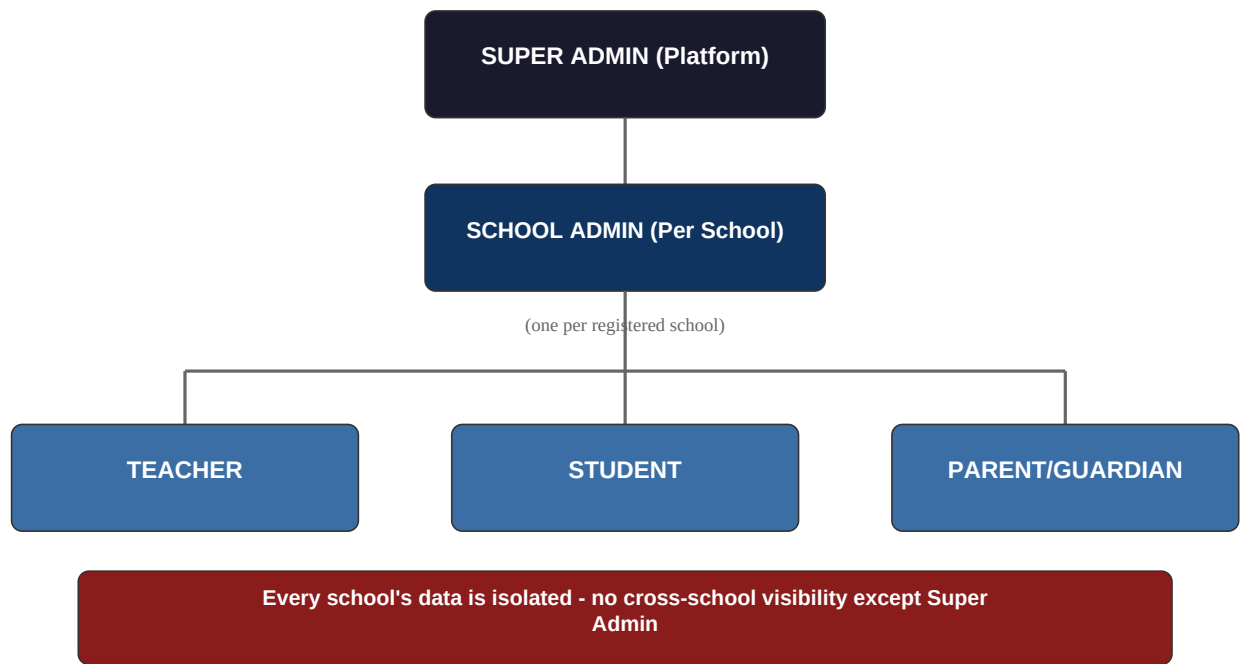
Role Structure

The platform is organized into two levels of administration, plus the day-to-day platform users:

Role	Scope	Purpose
Super Admin	Platform-wide	Oversees all registered schools, monitors activity, can suspend or reactivate a school account.
School Admin	One school	Manages that school's own setup: classes, subjects, teachers, students, parents, results, and announcements.
Teacher	One school	Manages assigned subjects/classes, enters results (added after Admin system is complete).
Student	One school	Accesses classes, assignments, results (added after Admin system is complete).
Parent/Guardian	One school	Views linked child(ren)'s progress and results (added after Admin system is complete).

Build order: the Admin system (Super Admin + School Admin) is being built first, since Teacher, Student, and Parent modules all depend on the structures Admin sets up (classes, subjects, accounts).

Role Hierarchy (Visual)



Module 1: Super Admin (Platform Level)

- Dashboard: total schools, total users, recent signups, activity trends.
- School directory: list and search all registered schools, view status (active/suspended).
- Suspend or reactivate a school account.
- Optional: view-only access into a school's data for support purposes.
- Optional: platform-wide announcements to all School Admins.

Module 2: School Signup & Onboarding

- Public registration form: school name, address, contact email, phone, school type.
- Auto-generated unique school ID created on signup, used to scope all future data.
- Email verification required before first login.
- First registrant is automatically granted the School Admin role.
- Onboarding checklist shown on first login: set up classes, add subjects, add teachers, add students.

Module 3: School Admin (Per-School Level)

School Profile

- School name, logo, address, contact info.
- Academic session/term settings (e.g., 2026/2027, First/Second/Third Term).

Class & Arm Management

- Class name (e.g., JSS1, SS2), arm (A, B, Science, Art).
- Class teacher assignment and student capacity.
- Create, edit, and delete classes and arms per session.

Subject Management

- Subject name, subject code, assigned class(es) and teacher(s).
- Core vs. elective subject flag for SS tracks (Science, Art, Commercial).

Teacher Management

- Full name, email, phone, assigned subjects and classes, employment status.
- Admin creates teacher accounts; login credentials sent by email.

Student Management

- Full name, admission number (auto-generated), class/arm, admission date, gender, linked parent/guardian.
- Bulk upload via CSV for adding many students at once.
- Promotion/repeat action at the end of term or session.

Parent/Guardian Linking

- Full name, email, phone, linked student(s).
- One parent account can link to multiple children within the same school.

Result / Report Card Oversight

- Admin reviews and approves results entered by teachers before release.
- Report card template settings: school logo, grading scale, comments section.

Communication

- Announcements visible only to that school's own teachers, students, and parents.
- Automated email triggers, e.g. account creation, result release.

Critical requirement: every record - class, subject, student, teacher, parent, result - must be scoped to its owning school's ID. No school should ever be able to view or access another school's data, with the sole exception of Super Admin oversight access.

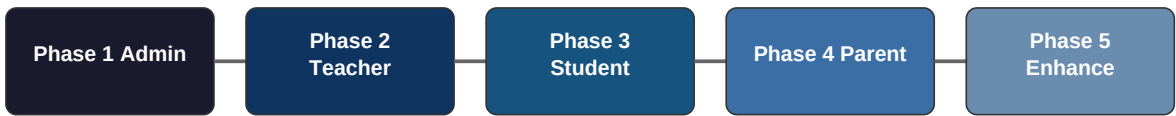
Phased Roadmap

The full platform is being built in phases, with each phase depending on the structures set up in the one before it.

Phase	Focus
Phase 1	Admin System - Super Admin + School Admin (current starting point)
Phase 2	Teacher Portal
Phase 3	Student Portal
Phase 4	Parent Portal
Phase 5	Enhancements

Each phase builds on the one before it - Teacher, Student, and Parent portals all depend on structures (classes, subjects, accounts) that Admin sets up first. Detail on each phase below.

Phase Flow (Visual)



Phase 2: Teacher Portal

Teachers log in to a school-scoped account created for them by the School Admin. Their view is limited to the subjects and classes assigned to them.

Dashboard

- Overview of assigned subjects, classes, and upcoming tasks (attendance to take, results to submit).

Attendance

- Mark attendance per class per day.
- View attendance history for a class.

Results & Continuous Assessment

- Enter test/CA scores and exam scores per subject per student.
- Submit results for Admin review before they are released to students/parents.

Assignments

- Post assignments to a class/subject with due dates.
- Review and grade submissions.

Class Materials

- Upload notes or resources for a specific subject/class (ties into the digital library concept).

Phase 3: Student Portal

Students log in to a school-scoped account, tied to their class/arm.

Dashboard

- Overview of timetable, upcoming assignments, and recent announcements.

Results / Report Card

- View term results once released by Admin.
- View cumulative/session performance.

Assignments

- View and submit assignments per subject.

Digital Library & Past Questions

- Access subject materials uploaded by teachers.
- Access past questions repository for exam preparation (flagged earlier as especially relevant for secondary schools).

Timetable

- View class timetable for the current term.

Phase 4: Parent Portal

Parents log in to an account linked to one or more students within the same school.

Dashboard

- Overview per linked child - attendance summary, recent results, announcements.

Results

- View each linked child's term results and report card once released.

Attendance

- View each linked child's attendance record.

Communication

- Receive school announcements and automated notifications (e.g., result released, fee due if in scope).

Phase 5: Enhancements (Post-Core)

Once Admin, Teacher, Student, and Parent are stable, these are worth considering as later additions - not part of the initial build.

- Fee tracking and payment records.
- Virtual/online classes.
- SMS notifications alongside email.
- Subscription/pricing tiers for schools, if the platform is meant to be commercially sustainable.

Infrastructure

The platform will build on the same free-tier stack used for the university system: Vercel for hosting the frontend, Render for backend services, and MongoDB Atlas for the database. Whether this is a new codebase/repository or an extension of the existing one is an open question below.

Ideas to Consider (Open for Discussion)

A few suggestions worth discussing as the build progresses - none of these are decided yet, just flagged for your input:

- Subscription/pricing tiers for schools (e.g., free tier with limits, paid tier for larger schools) if this is meant to be commercially sustainable.
- A super admin support-access mode, so issues can be diagnosed for a school without needing their login.
- Audit logging on admin actions (who created/edited/deleted a record), useful once multiple schools and staff are on the platform.
- Whether this is a new codebase/repository or an extension of the existing university system's codebase.
- Starting Phase 1 with a single pilot school's data even though the architecture is multi-tenant from day one, to validate the flow before opening signup publicly.
- A rough timeline/effort estimate per phase, once scope is agreed - not fixed dates, just a sense of scale.

Before We Start Building

This document is a starting point for a conversation, not a finished spec. The goal is to go through it together, agree on scope for Phase 1, and settle the open questions above before any coding begins.

- Read through and flag anything that doesn't match your understanding of the project.
- Talk through the open questions - especially what carries over from the university system and codebase direction.
- Agree on the exact scope of Phase 1 (Admin System) as the starting point.
- Only after that's settled, move into detailed build planning for Phase 1.